A-Level Chemistry Worksheet

The Haber Process: History and Chemical Principles

Name: ____________________	Date: ____________________	Total: _____ / 35

Background context

The Haber Process is the industrial synthesis of ammonia (NH_3) from nitrogen and hydrogen. Before it existed, fixed nitrogen was scarce, sourced mainly from Chilean sodium nitrate deposits.

In 1909, German chemist **Fritz Haber** demonstrated laboratory synthesis of ammonia at ~550 °C and ~175 atm using an osmium catalyst. **Carl Bosch** of BASF then scaled the process industrially by 1913, developing high-pressure vessels and a promoted iron catalyst.

The overall equation is: $\mathbf{N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g) \quad \Delta H = -92\ kJ\ mol^{-1}}$

Industrial conditions: ~450 °C, ~200 atm, iron catalyst promoted with K_2O and Al_2O_3. Typical yield: ~15 %. Both Haber (1918) and Bosch (1931) received Nobel Prizes.

Section A: History of the Haber Process [10 marks]

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1. Name the German chemist who first demonstrated the laboratory synthesis of ammonia. State the year this was achieved. *[2 marks]*

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2. Describe the role of Carl Bosch in the development of the Haber Process. *[2 marks]*

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3. The iron catalyst in the Haber Process is used with promoters. Name one promoter and explain its function. *[2 marks]*

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4. Give **one** wartime application and **one** modern peacetime application of the ammonia produced by the Haber Process. *[2 marks]*

(a) Wartime: ____

(b) Modern: ____

5. Fritz Haber received the Nobel Prize in Chemistry in 1918. Suggest why this award was considered controversial. *[2 marks]*

Section B: Multiple Choice — Chemical Principles [8 marks]

Circle the letter of the correct answer for each question.

B1. What is the total number of moles of **gas** on the **left-hand side** of the Haber Process equation? *[2 marks]*

- **A.** 1
- **B.** 2
- **C.** 3
- **D.** 4

B2. Increasing the pressure in the Haber Process causes the equilibrium position to: *[2 marks]*

- **A.** Shift left, favouring decomposition of NH_3
- **B.** Shift right, favouring formation of NH_3
- **C.** Remain unchanged — pressure has no effect on equilibrium position
- **D.** Shift right only if temperature is simultaneously decreased

B3. The industrial process operates at ~450 °C rather than a lower temperature. The main reason for this is: *[2 marks]*

- **A.** The iron catalyst only functions above 400 °C
- **B.** Lower temperatures give a higher equilibrium yield but an unacceptably slow reaction rate
- **C.** The forward reaction is endothermic, so higher temperature shifts equilibrium to the right
- **D.** Lower temperatures cause the reactant gases to liquefy

B4. Which statement correctly describes the catalyst used in the Haber Process and its effect? *[2 marks]*

- **A.** Platinum; increases the equilibrium yield of NH_3
- **B.** Iron; increases both the rate of reaction and the equilibrium yield of NH_3
- **C.** Iron; increases the rate of reaction without altering the equilibrium position
- **D.** Vanadium(V) oxide; increases the rate without altering the equilibrium position

Section C: Diagrams and Labelling [8 marks]

C1. In the axes below, sketch concentration–time curves for N_2, H_2, and NH_3 as the Haber Process reaches equilibrium. Label each curve, mark the point at which equilibrium is established, and add appropriate axis labels. *[4 marks]*

Concentration–time graph (label axes, curves, and equilibrium point)

C2. Sketch graphs showing how equilibrium percentage yield of NH_3 varies with (a) temperature at constant pressure, and (b) pressure at constant temperature. Label axes and indicate the trend. *[4 marks]*

(a) % yield NH_3 vs Temperature (constant P)	**(b) % yield NH_3 vs Pressure (constant T)**

Section D: Extended Writing [9 marks]

D1. Evaluate the choice of operating conditions in the industrial Haber Process — temperature (~450 °C), pressure (~200 atm), and the iron catalyst. In your answer: *[9 marks]*

- explain the effect of each condition on equilibrium yield and reaction rate
- apply Le Chatelier's Principle to justify each choice
- discuss the economic and practical trade-offs involved

You should write in continuous prose. A well-structured answer will address all three conditions and link chemical principles to economic reasoning.